

1. "Meaningful questions about the dataset before analysis."

i. Which books are most expensive?

ii. What rating appears most frequently?

iii. Are expensive books highly rated?

iv. How many books are in stock?

```
In [4]: import pandas as pd  
import numpy as np
```

```
In [5]: data=pd.read_csv("books_dataset.csv")
```

```
In [6]: data
```

Out[6]:

	Title	Price	Rating	Price_GBP	Price_INR	Stock
0	A Light in the Attic	£51.77	Three	51.77	5435.85	In stock
1	Tipping the Velvet	£53.74	One	53.74	5642.70	In stock
2	Soumission	£50.10	One	50.10	5260.50	In stock
3	Sharp Objects	£47.82	Four	47.82	5021.10	In stock
4	Sapiens: A Brief History of Humankind	£54.23	Five	54.23	5694.15	In stock
...	...	...	...	...	...	...
995	Alice in Wonderland (Alice's Adventures in Won...	£55.53	One	55.53	5830.65	In stock
996	Ajin: Demi-Human, Volume 1 (Ajin: Demi-Human #1)	£57.06	Four	57.06	5991.30	In stock
997	A Spy's Devotion (The Regency Spies of London #1)	£16.97	Five	16.97	1781.85	In stock
998	1st to Die (Women's Murder Club #1)	£53.98	One	53.98	5667.90	In stock
999	1,000 Places to See Before You Die	£26.08	Five	26.08	2738.40	In stock

1000 rows × 6 columns

In [7]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Title       1000 non-null   object
1   Price       1000 non-null   object
2   Rating      1000 non-null   object
3   Price_GBP   1000 non-null   float64
4   Price_INR   1000 non-null   float64
5   Stock       1000 non-null   object
dtypes: float64(2), object(4)
memory usage: 47.0+ KB
```

In [13]: `data.drop("Price",axis=1,inplace=True)`

2. "Explore the data structure, including variables and data types."

In [14]: data

Out[14]:

	Title	Rating	Price_GBP	Price_INR	Stock
0	A Light in the Attic	Three	51.77	5435.85	In stock
1	Tipping the Velvet	One	53.74	5642.70	In stock
2	Soumission	One	50.10	5260.50	In stock
3	Sharp Objects	Four	47.82	5021.10	In stock
4	Sapiens: A Brief History of Humankind	Five	54.23	5694.15	In stock
...	...	...	...	...	...
995	Alice in Wonderland (Alice's Adventures in Won...	One	55.53	5830.65	In stock
996	Ajin: Demi-Human, Volume 1 (Ajin: Demi-Human #1)	Four	57.06	5991.30	In stock
997	A Spy's Devotion (The Regency Spies of London #1)	Five	16.97	1781.85	In stock
998	1st to Die (Women's Murder Club #1)	One	53.98	5667.90	In stock
999	1,000 Places to See Before You Die	Five	26.08	2738.40	In stock

1000 rows × 5 columns

In [16]: data.shape

Out[16]: (1000, 5)

In [17]: data.dtypes

```
Out[17]: Title      object
Rating    object
Price_GBP float64
Price_INR float64
Stock     object
dtype: object
```

```
In [18]: data.head()
```

```
Out[18]:
```

	Title	Rating	Price_GBP	Price_INR	Stock
0	A Light in the Attic	Three	51.77	5435.85	In stock
1	Tipping the Velvet	One	53.74	5642.70	In stock
2	Soumission	One	50.10	5260.50	In stock
3	Sharp Objects	Four	47.82	5021.10	In stock
4	Sapiens: A Brief History of Humankind	Five	54.23	5694.15	In stock

```
In [21]: data.isnull().sum()
```

```
Out[21]: Title      0
Rating    0
Price_GBP  0
Price_INR  0
Stock     0
dtype: int64
```

3. "Identify trends, patterns and anomalies within the data."

```
In [19]: data["Rating"].value_counts()
```

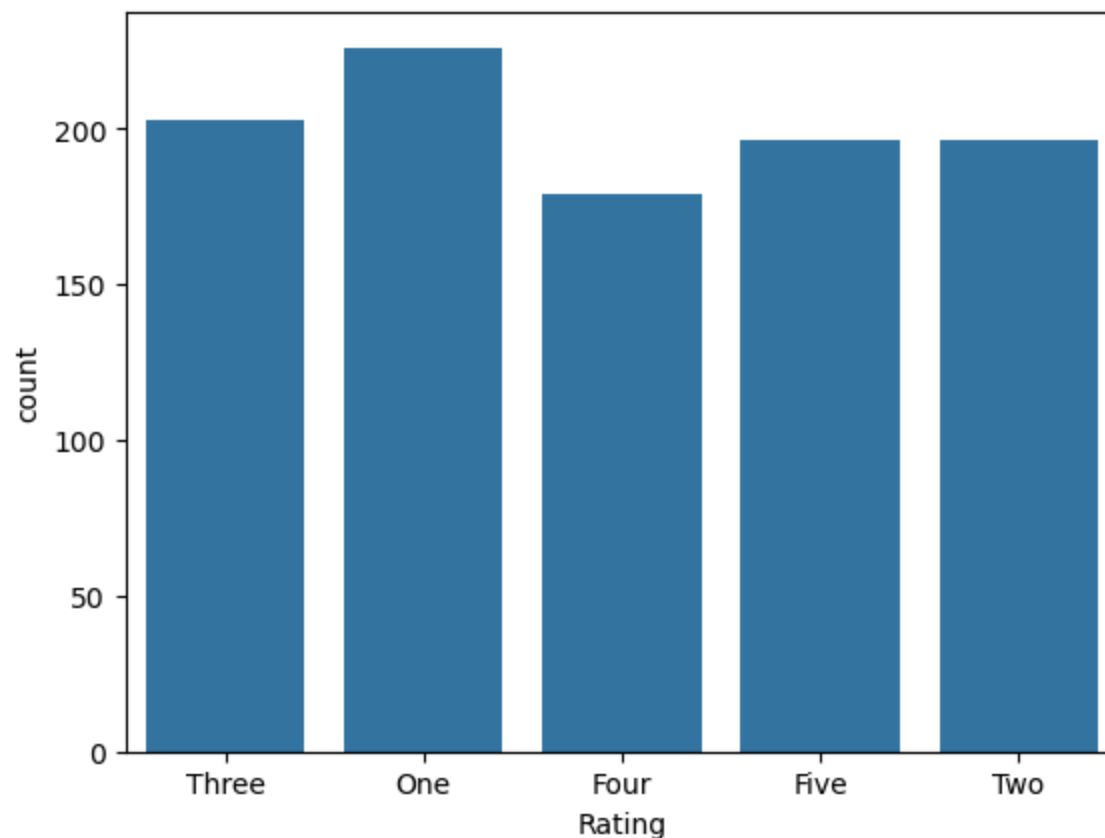
```
Out[19]: Rating
         One      226
         Three    203
         Five     196
         Two      196
         Four     179
         Name: count, dtype: int64
```

```
In [20]: data['Rating'].describe()
```

```
Out[20]: count      1000
         unique        5
         top          One
         freq         226
         Name: Rating, dtype: object
```

```
In [23]: import seaborn as sns
         import matplotlib.pyplot as plt
```

```
In [24]: sns.countplot(x="Rating", data=data)
         plt.show()
```



4. Test hypotheses and validate assumptions using statistics and visualization

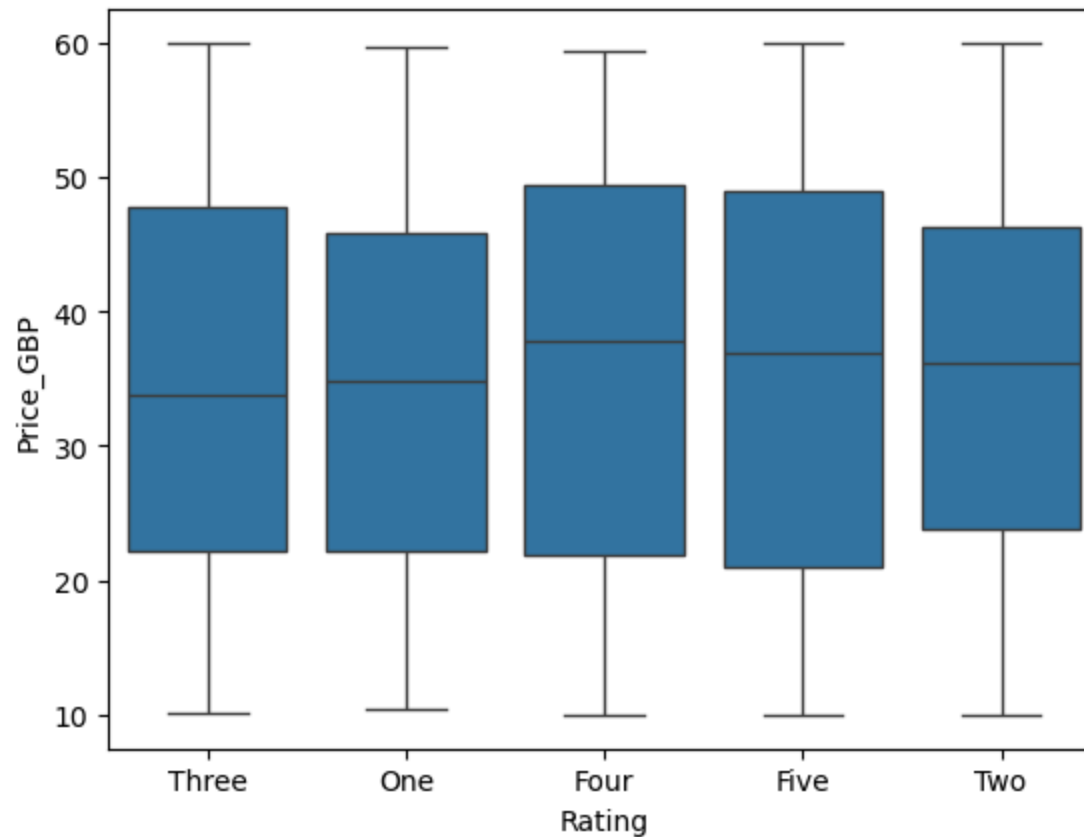
```
In [25]: data.groupby("Rating")["Price_GBP"].mean()
```

```
Out[25]: Rating
Five      35.374490
Four      36.093296
One       34.561195
Three     34.692020
Two       34.810918
Name: Price_GBP, dtype: float64
```

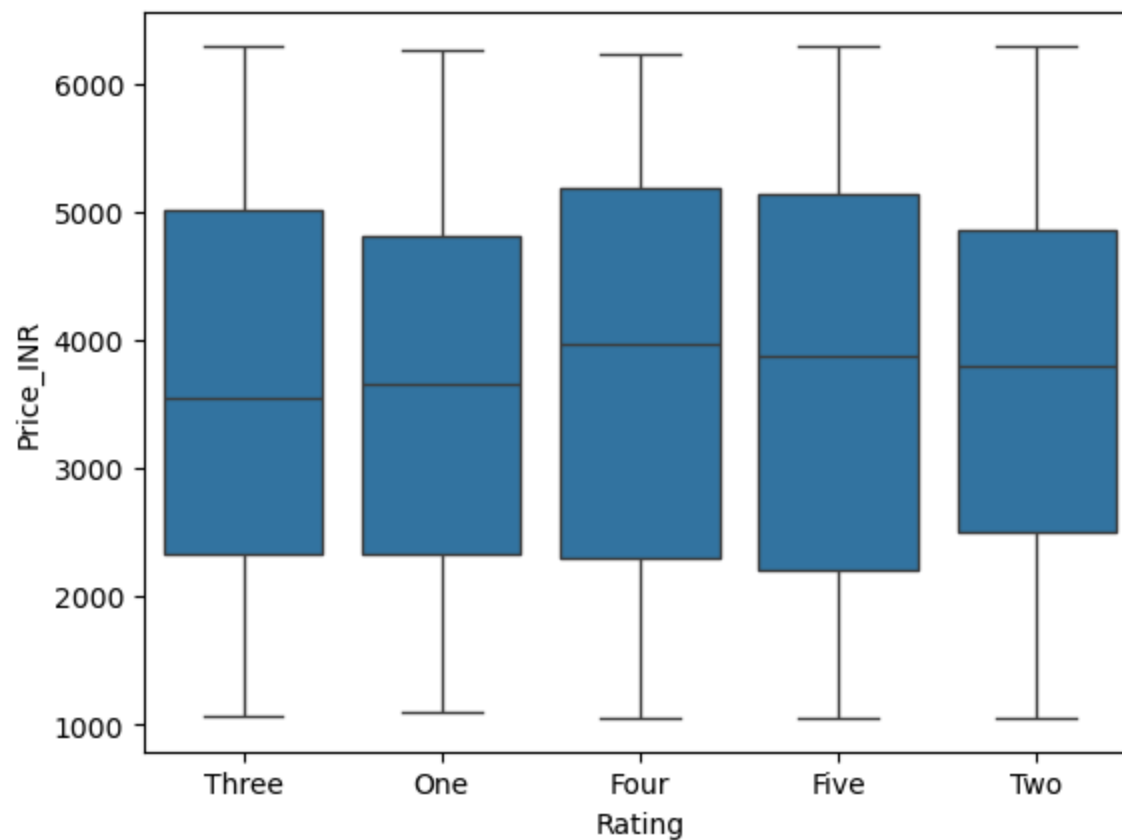
```
In [26]: data.groupby("Rating")["Price_INR"].mean()
```

```
Out[26]: Rating
Five      3714.321429
Four      3789.796089
One       3628.925442
Three     3642.662069
Two       3655.146429
Name: Price_INR, dtype: float64
```

```
In [27]: sns.boxplot(x="Rating", y="Price_GBP", data=data)
plt.show()
```



```
In [29]: sns.boxplot(x="Rating", y="Price_INR", data=data)
plt.show()
```



5. "Detect potential data issues or problems to address in further analysis."

```
In [32]: data.duplicated().sum()
```

```
Out[32]: np.int64(0)
```

```
In [33]: data.info()
```



```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1000 entries, 0 to 999  
Data columns (total 5 columns):  
#   Column      Non-Null Count  Dtype  
---  ---  
0   Title       1000 non-null   object  
1   Rating      1000 non-null   object  
2   Price_GBP   1000 non-null   float64  
3   Price_INR   1000 non-null   float64  
4   Stock       1000 non-null   object  
dtypes: float64(2), object(3)  
memory usage: 39.2+ KB
```

```
In [34]: data.isnull().sum()
```

```
Out[34]: Title      0  
Rating    0  
Price_GBP  0  
Price_INR  0  
Stock     0  
dtype: int64
```

Answer The Questions

i. Which books are most expensive?

```
In [35]: data['Price_INR'].max()
```

```
Out[35]: 6298.95
```

```
In [40]: data.sort_values(by="Price_GBP", ascending=False).head()
```

Out[40]:

	Title	Rating	Price_GBP	Price_INR	Stock
648	The Perfect Play (Play by Play #1)	Three	59.99	6298.95	In stock
617	Last One Home (New Beginnings #1)	Three	59.98	6297.90	In stock
860	Civilization and Its Discontents	Two	59.95	6294.75	In stock
560	The Barefoot Contessa Cookbook	Five	59.92	6291.60	In stock
366	The Diary of a Young Girl	Three	59.90	6289.50	In stock

ii. What rating appears most frequently?

In [41]: `data['Rating'].value_counts()`

Out[41]:

Rating	
One	226
Three	203
Five	196
Two	196
Four	179

Name: count, dtype: int64

iii. Are expensive books highly rated?

In [43]: `data.groupby("Rating")["Price_GBP"].mean()`

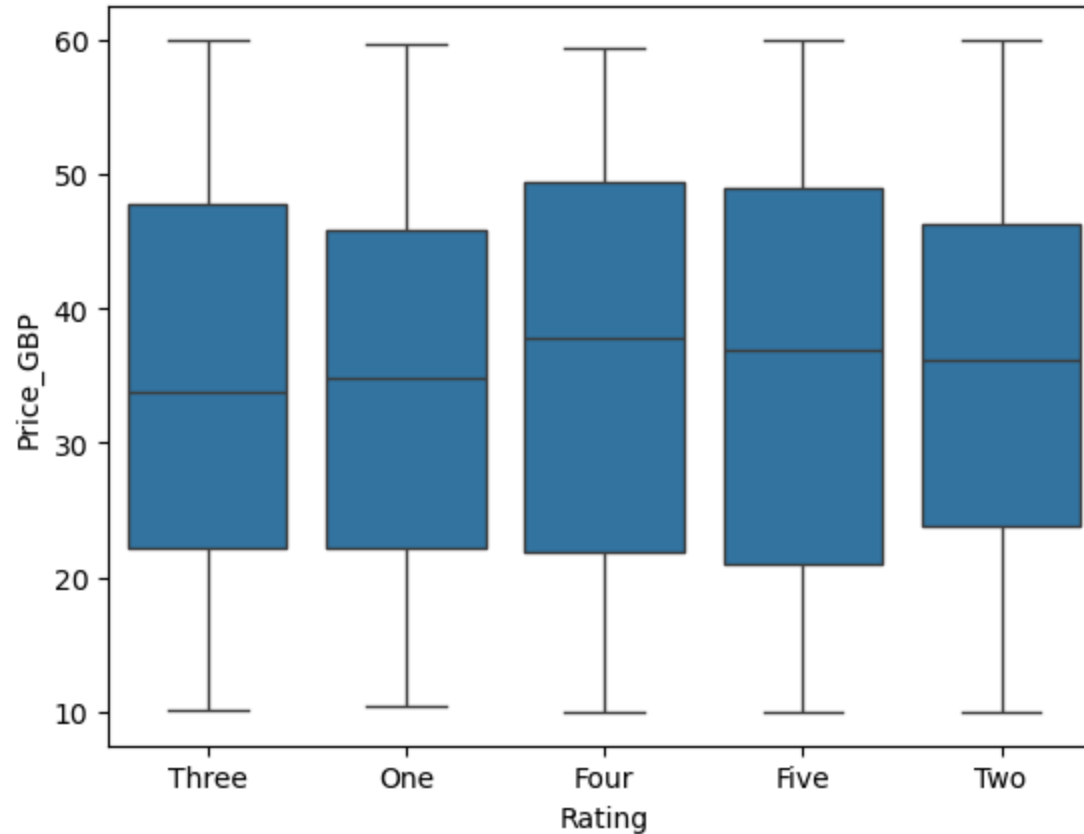
Out[43]:

Rating	
Five	35.374490
Four	36.093296
One	34.561195
Three	34.692020
Two	34.810918

Name: Price_GBP, dtype: float64

In [44]: `sns.boxplot(x="Rating", y="Price_GBP", data=data)`

```
plt.show()
```



iv. How many books are in stock?

```
In [45]: data['Stock'].value_counts()
```

```
Out[45]: Stock  
In stock    1000  
Name: count, dtype: int64
```

Summary of Exploratory Data Analysis

Based on the analysis of the book dataset collected from Books to Scrape, the following insights were observed:

```
In [ ]: 1 Most Expensive Books
The dataset contains several books with higher prices, with the most expensive books priced approximately between £50 a

2 Most Frequent Rating
The analysis of the rating distribution shows that 1-star ratings appear most frequently in the dataset.

3 Relationship Between Price and Rating
The comparison between book prices and ratings suggests that higher-rated books tend to have slightly higher average pr

4 Stock Availability
The dataset indicates that all books are currently listed as "In stock."
```